Computer Methods and Programs in Biomedicine xxx (xxxx) xxx

Contents lists available at ScienceDirect

Computer Methods and Programs in Biomedicine

journal homepage: www.elsevier.com/locate/cmpb

Current Application of Digital Diagnosing Systems for Retinopathy of Prematurity

Yuekun Bao[a,g], Wai-Kit Ming[b], Zhi-Wei Mou[c], Qi-Hang Kong[a], Ang Li[d,e,*], Ti-Fei Yuan[f,*], Xue-Song Mi[a,h,*]

[a] Department of Ophthalmology, the First Affiliated Hospital of Jinan University, Guangzhou, China
[b] Clinical Medicine, International School, Jinan University, Guangzhou, China
[c] Department of Rehabilitation, the First Affiliated Hospital of Jinan University, Guangzhou, China
[d] Guangdong - Hong Kong - Macau Institute of CNS Regeneration, Joint International Research Laboratory of CNS Regeneration Ministry of Education, Jinan University, Guangzhou, China
[e] Bioland Laboratory (Guangzhou Regenerative Medicine and Health Guangdong Laboratory), Guangzhou, China
[f] Shanghai Key Laboratory of Psychotic Disorders, Shanghai Mental Health Center, Shanghai Jiaotong University School of Medicine, Shanghai, China
[g] State Key Laboratory of Ophthalmology, Zhongshan Ophthalmic Center, Sun Yat-sen University, Guangzhou, China
[h] Changsha Academician Expert Workstation, Aier Eye Hospital Group, Changsha, China

ARTICLE INFO

Article history:
Received 30 March 2020
Accepted 18 November 2020
Available online xxx

Keywords:
retinopathy of prematurity
computer-based image analysis
machine learning
deep learning
multiple instance learning

ABSTRACT

Background and Objective: Retinopathy of prematurity (ROP), a proliferative vascular eye disease, is one of the leading causes of blindness in childhood and prevails in premature infants with low-birth-weight. The recent progress in digital image analysis offers novel strategies for ROP diagnosis. This paper provides a comprehensive review on the development of digital diagnosing systems for ROP to software researchers. It may also be adopted as a guide to ophthalmologists for selecting the most suitable diagnostic software in the clinical setting, particularly for the remote ophthalmic support.

Methods: We review the latest literatures concerning the application of digital diagnosing systems for ROP. The diagnosing systems are analyzed and categorized. Articles published between 1998 and 2020 were screened with the two searching engines Pubmed and Google Scholar.

Results: Telemedicine is a method of remote image interpretation that can provide medical service to remote regions, and yet requires training to local operators. On the basis of image collection in telemedicine, computer-based image analytical systems for ROP were later developed. So far, the aforementioned systems have been mainly developed by virtue of classic machine learning, deep learning (DL) and multiple machine learning. During the past two decades, various computer-aided systems for ROP based on classic machine learning (e.g. RISA, ROPtool, CAIER) became available and have achieved satisfactory performance. Further, automated systems for ROP diagnosis based on DL are developed for clinical applications and exhibit high accuracy. Moreover, multiple instance learning is another method to establish an automated system for ROP detection besides DL, which, however, warrants further investigation in future.

Conclusion: At present, the incorporation of computer-based image analysis with telemedicine potentially enables the detection, supervision and in-time treatment of ROP for the preterm babies.

© 2020 Elsevier B.V. All rights reserved.

1. Introduction

Retinopathy of prematurity (ROP) is a vascular disease prevalent in premature infants, especially those with a birth weight lower than 1250 g. As a major cause of blindness in infants and young children [1], the incidence of ROP is rising due to the rapid improvement of rescue technology in neonatology worldwide. Nonetheless, the course of ROP is short, thereby making the effective therapeutic window relatively narrow. Once the disease progresses to the late stage or the tractional retinal detachment occurs, the prognosis of visual functions will be poor. Hence, diagnosing and treating ROP timely and accurately can effectively

* Correspondence authors.
E-mail addresses: anglijnu@jnu.edu.cn (A. Li), ytf0707@126.com (T.-F. Yuan), mixuesong_eye@163.com (X.-S. Mi).

https://doi.org/10.1016/j.cmpb.2020.105871
0169-2607/© 2020 Elsevier B.V. All rights reserved.

Please cite this article as: Y. Bao, W.-K. Ming, Z.-W. Mou et al., Current Application of Digital Diagnosing Systems for Retinopathy of Prematurity, Computer Methods and Programs in Biomedicine, https://doi.org/10.1016/j.cmpb.2020.105871

prevent blindness. According to the Early Treatment for Retinopathy of Prematurity (ETROP) trial [2], a timely clinical intervention is conducive to enhance the vision of high-risk patients with ROP, even though 9% became blind ultimately. Thus, the early screening and regular monitoring are particularly crucial. Plus disease in ROP is a key feature that requires treatment and can be identified by comparison with the published photographs regarding the arterial tortuosity and venous dilation of posterior retinal vessels. Therefore, conducting research on time- and labor-saving screening for ROP plus disease is necessary. Current therapies for ROP mainly include laser photocoagulation, intraocular injection, cryotherapy, scleral buckling and vitrectomy, based on the stages of the disease. Still, timely treatments are the key point for a good prognosis, given the fact that in the late stage of ROP, maneuvers like vitrectomy can manage to reduce a partial or complete anatomical impairment and yet fail to retain the normal visual function. Accordingly, it is critical to identify the treatment-warranted ROP (TW-ROP) in the high-risk infants.

In recent years, numerous morphological datasets have become accessible by digital images. By use of these large databases, retinal images can be analyzed efficiently with artificial intelligence (AI). Methods involving machine learning (ML) especially deep learning (DL) are capable of recognizing pathological characteristics in retinal diseases. Automated AI programs have been well developed and widely applied to screen diabetes in many countries. A breakthrough is that the world's first market-oriented AI medical device developed by Abramoff *et al*. was approved by the United States Food and Drug Administration (US FDA) in the April, 2018, which allowed automatic screening and diagnosis of diabetic retinopathy (DR) in diabetic adults using the autonomous DR and DME detection software (IDx-DR) [3]. The progresses in computer-aided technology are also gradually changing the diagnostic method for ROP. Although the traditional binocular indirect ophthalmoscopy (BIO) remains the golden standard for ROP screening, it requests the direct practice by experienced clinicians. The wide-angle digital retinal imaging system (Retcam) is increasingly utilized for fundus examination of ROP nowadays. Compared with BIO, Retcam is easier to operate with a shorter learning cycle, and in the meantime can be used for photographing, storage, output and transmission of multi-directional fundus images. Further, Retcam system is more conducive for follow-up, consultation, scientific research and education. These advantages of Retcam guarantee satisfactory results when applying it to diagnostic digital image analysis on ROP.

Telemedicine is a method of long-distance image interpretation, which provides medical service to remote regions but requires training to the local operators. Developed from the concept of image collection and analysis in telemedicine, various systems of computer-based image analysis using classic ML have been invented with promising outcomes. These systems were developed based on a small number of images and have relatively low demand for web server. However, they require manual delineation of selected features, and thus systemic bias may result from handcrafted extraction. Automated systems for ROP diagnosis based on DL have been developed for clinical applications. These systems are more time- and labor-saving compared with those based on classic ML. Unfortunately, they still call for large datasets covering various ROP features and different severities, together with high demand for web server. Notably, for convolutional neural networks, innovative programs are expected to be designed based on different datasets with the population from distinct countries, so that the networks can be modified properly in various situations. Moreover, it is difficult to compare the existing systems in a uniform way owing to the lack of a standard strategy. Consequently, it is called upon to establish a program able to compare the existing ROP programs. Also, multiple instance learning (MIL) is anticipated to be applicable for automated diagnosis in future.

In this work, we have reviewed the current application of digital diagnosing systems for ROP by use of classic ML methods, MIL and DL. We compared these systems with analysis parameters, outcomes, performances, *etc* (Table 1). We found that to date, each currently-used system has its own merits and disadvantages. An incorporation of telemedicine and computer-based image analysis potentially enhances detection, supervision and in-time treatment of ROP for the preterm babies.

2. Methodology

We conducted a comprehensive review of published literatures on the application of diagnosing systems for ROP. We developed a search strategy in consultation with a senior reviewer, and searched through the online engines PubMed and Google Scholar for articles containing the following terms: retinopathy of prematurity, artificial intelligence, digital diagnosing systems, deep learning, computer-based image analysis, machine learning, and telemedicine. We performed the searching with MeSH terms (retinopathy of prematurity [Title/Abstract]) AND (digital diagnosing systems [Title/Abstract] OR computer-based image analysis [Title/Abstract] OR machine learning [Title/Abstract] OR deep learning [Title/Abstract] OR telemedicine [Title/Abstract] OR artificial intelligence [Title/Abstract]). We set the filter to select the original articles published in the past ten years which are in full text and written in English, resulting 148 papers. The papers were then reviewed for subject relevance via the abstracts of the articles. For example, papers investigating pediatric ophthalmology and artificial intelligence were excluded from review. We also searched the reference lists of any articles selected for inclusion in this review in order to identify additional relevant articles.

3. Results

3.1. Telemedicine

As a method for long-range image interpretation, telemedicine offers patients in remote regions a chance of better medical service, despite that the final quality depends highly on local operators. In 2013, the application of digital retinal image photographed for remote interpretation of ROP screening was acknowledged by the American Academy Of Pediatrics (AAP), American Academy of Ophthalmology (AAO), and American Association for Pediatric Ophthalmology and Strabismus (AAPOS) for the first time [4]. The study designs of existing telemedicine programs comprise 5 aspects, namely (1) the number of wide-angle retinal images, (2) the camera operators including ophthalmologists [5], ophthalmic operators [6] and trained NICU nurses [7,8], (3) the image graders including ROP specialists [6,8], pediatric ophthalmologists [5,9] and general ophthalmologists [6], (4) the diagnostic outcomes including moderate ROP (e.g. the presence of Type 2 or more types of diseases) [5,6] and severe ROP (e.g. the presence of treatment-requiring diseases) [8,10], and (5) the experimental results. Some computer-based image analysis (CBIA) studies relied on the images collected in telemedicine programs [11], such as KIDROP, India's largest telemedicine initiative [12]. A large-scale retrospective study of telemedicine for ROP, known as the 6-year result of Stanford University Network for Diagnosis of Retinopathy of Prematurity (SUNDROP), was reported in 2015 [8], where retinal photographs were evaluated by the ROP specialists, and the golden standard was the clinical diagnosis from the bedside BIO performed by a pediatric retinal specialist; TW-ROP and other anatomic findings were the primary outcomes. Compared with the bedside BIO, the remote image interpretation led to the diagnosis results with the sensitivity of 100%, specificity of 99.8%, PPV of 95.5%, and NPV of 100%. Taken together, telemedicine appears as

Table 1
Summary of different digital diagnosing systems for ROP

Method	Type of ML	Outcome	No. of images	FOV	Device	Performance	Ref.
ROPtool system	CML	pre-plus & plus	20	45°	RetCam	Pictorial scales correlated well with tortuosity and dilation indices.	[21]
Gabor filters	CML	plus & non-plus	19 & 91	130°	RetCam	Tortuosity: 89% sensitivity, 95% specificity; Vessel thickness: 73% sensitivity, 66% specificity.	[26,27]
NEAT-O system	CML	plus & non-plus	N/A	2.5DD	RetCam	Clinical assessments strongly correlated with the golden standard; Pearson's correlation coefficient was 0.802.	[23]
Gaussian matched filter	CML	APROP & normal	21 & 15	130°	RetCam	100% recall and 95.45% precision in morphologically obtained vessel networks.	[12]
MLP classifier	CML	pre-plus & plus & non-plus	12 & 38 & 37	130°	RetCam	Accuracy: 72.3% for plus, 83.7% for pre-plus, and 84.45 for non-plus.	[22]
i-ROP system	CML	normal & pre-plus & plus	47 & 16 & 14	6DD	RetCam	Accuracy: 95%; Sensitivity: 97% for pre-plus or worse, 93% for plus.	[28]
i-ROP DL system	DL	normal & pre-plus & plus	4535 & 805 & 172	PP	RetCam	Sensitivity: 93% for plus, 100% for pre-plus or worse; Specificity: 94% for plus, 94% for pre-plus or worse.	[35]
i-ROP DL system	DL	VSS	5255	PP	RetCam	Median severity scores: 1.1 for no ROP, 1.5 for mild ROP, 4.6 for Type 2 & pre-plus, and 7.5 for ROP requiring treatment.	[36]
i-ROP DL system	DL	VSS	5255	PP	RetCam	Mean severity scores: before treatment: 4.19; at the time of treatment: 7.43; after treatment: 4.00.	[44]
DeepROP system	DL	normal & minor & severe	7559 & 1834 & 2305	N/A	RetCam	Id-Net: 96.64% sensitivity, 99.33% specificity; Gr-Net: 88.46% sensitivity, 92.31% specificity.	[45]
ROP.AI system	DL	normal & plus	3792 & 1134	130°	RetCam	96.6% sensitivity, 98.0% specificity.	[46]
miGraph classifer	MIL	healthy & ROP	28 & 28	130°	RetCam	Accuracy >95%.	[53]

Note: ROP=retinopathy of prematurity; ML=machine learning; FOV=field of view; CML=classic ML; DL=deep learning; MIL=multiple instance learning; APROP=aggressive posterior ROP; VSS=vascular severity score; DD=disc diameter; PP=posterior pole; N/A=not available.

a safe, reliable and cost-effective auxiliary tool for ROP specialists, particularly for the remote ophthalmic support.

3.1.1. Advantages and limitations

Telemedicine for ROP has a great potential to improve the efficiency in disease prevention and management. It owns multiple merits including that patients can get access to local high-quality medical services and that ophthalmologists are able to spend much less time in commuting to NICUs and waiting for dilation of pupils. Besides, the imbalanced distribution of medical resources in urban and rural regions can also be reduced [13]. So far, there have been numerous successful examples of telemedicine application in the screen and diagnosis of ROP, all of which displayed the high accuracy, reliability and cost-effectiveness [14]. Nevertheless, the remote retinal image screening through telemedicine requires considerable upfront costs (e.g. equipment purchase) and the subsequent standardization in applications [15], such as establishing the standard of technical regulations with operational procedures and completing the consistency of the diagnostic and referral standards [16]. Some concerns of digital imaging analysis and remote image interpretation are described in the statement on screening of preterm infants for ROP published in 2015 [4]. For example, technical concerns may be related to the training and personnel of both photographers and image analyzers. It is also essential to improve the personnel training and emphasize the medical liability in advance. Moreover, it is important to enhance the management of data security and protect the privacy of patients [17].

The quality of the photos is also worthy of concern since there are many influencing factors from the capture of a photo to its evaluation, such as different photographers, varied fields of view required for interpretation, and so forth. Fortunately, a couple of telemedicine studies have reported that the accuracy of images taken by non-ophthalmologists and by ophthalmologists is comparable [17].

3.2. Current development of computer-based image analysis (CBIA)

As mentioned above, CBIA systems for diagnosing ROP were developed from the concept of image collection in telemedicine, and the current platforms have mostly relied on the methods of classic ML, DL and multiple machine learning, as elaborated below. Parameter values of different methods are concluded in Supplementary Table 1.

3.2.1. Classic Machine Learning

During the past two decades, a variety of computer-aided systems, such as RISA, ROPtool [18,19] and CAIER, have been established for ROP, especially for detection of plus disease. Computer systems designed for analysis of the preterm retina have provided useful tools for vessel evaluation.

ROPtool evaluated tortuosity from 20 images based on the arch and chord length of vessels to classify plus or pre-plus disease patients, or normal cases [20]. A pilot study in 2018 using ROPtool created a pictorial scale of vascular tortuosity and dilation according to vascular characteristics distinguished by the specialists. The researchers collected narrow-field pictures to establish a simplified pictorial scale. The pictures were ordered based on the dilation and tortuosity. All the pictures were examined and diagnosed by three ROP experts. Tortuosity measured by ROPtool was the traced vessels divided by a smooth curve, and dilation was the average vessel width over its length divided by the area of the optic nerve. The dilation and tortuosity indices were calculated via multiplying the original value by an adjustment factor, and an index of 10 represented the minimal abnormality required for the plus disease, as defined by the representative photo. This pictorial scale could be employed as a reference to reduce subjectivity in ROP diagnosis [21]. Pour *et al.* used machine language program (MLP) classifier to develop algorithms with 38 plus, 12 pre-plus, and 37 non-plus images based on two methods. The researchers created artificial images of sinusoidal waveforms and lines in various diameters similar

to the fundus images at first. Thereafter, they applied it to the real images after having reasonable results. Two parameters of plus disease detection included the curvature measure and distance transform for assessment of tortuosity and vascular dilatation. This algorithm for diagnosis and screening of plus disease had acceptable accuracy [22]. However, due to issues like image quality, standardizing of image acquisition and time for image processing, such algorithms were rarely used in the clinical setting. Poor image quality often severely limits the segmentation in automated softwares.

The NEAT system for assessment of tortuosity using 50 individual vessels and 167 eyes to determine an overall severity of tortuosity was clinically instrumental in poor-quality images; this is different from other previous diagnosing systems [23]. First, the morphological opening procedure was performed to unify the background intensity of images in each color channels. Thereafter, every channel would go through an adaptive histogram equalization process. The individual color channels were connected back to a single image from which the final grayscale image was obtained. The processed image would finally be presented to the user to initiate blood vessel segmentation. NEAT system relied on mathematical topology techniques to integrate the degree of vascular curvature. The user highlighted each peak of the inflection point of the vessels and delineated those to be evaluated with a linear boundary. Other than tortuosity, vessel dilation and diameter were regarded as parameters in these studies. Several investigations have suggested that it was not important to distinguish the type of vessels because a significant correlation existed between treated and non-treated eyes based on the most tortuous vessel [24,25]. On the other hand, the selection of vessel branches can be a key factor in evaluation. Oloumi *et al.* used Gabor filters to detect the major temporal arcade (MTA), which is the thickest venular branch in the retina [26,27]. A total of 110 images of 41 patients were selected. Five different images were representative of different retinal areas. In each case, the most visible image of two quadrants of MTA was selected. The original photos were converted into Gabor-magnitude and angle-response images to extract the skeleton of MTA. If an image contained at least 2.5 mm of abnormally twisted vessels in each of at least 2 quadrants or at least 5 mm of abnormally twisted vessels in any of the quadrants, the diagnosis was plus disease. In plus disease, the mean and the median MTA widths were 15 mm and 22 mm greater than normal images, respectively. The results showed a significant difference in MTA width between the normal and plus disease cases, thereby indicating the measurement of MTA could assist in plus disease diagnosis. The classification of ROP was also a consideration according to some studies. Aggressive posterior retinopathy of prematurity (APROP) is characterized by severe plus disease, flat neovascularization in Zone 1 or posterior Zone 2, intraretinal shunting, hemorrhage and a rapid progression to retinal detachment. Rajashekar *et al.* used Gaussian matched filter to quantify vessel characteristics that were of clinical relevance to APROP, such as tortuosity and the extent of branching from 15 retinal images [12]. A total of 36 images were obtained, including 21 from APROP patients and 15 from healthy controls. There were three preprocessing steps: saturation noise, optic disc mask, and channel selection. Since APROP occurs close to the optic disc (OD), the area 4*radius OD (DR1) and 6*radius (EDR1) were focused to explore the vessel activity. Two parameters of APROP detection in the selected areas included the tortuosity index and the segment count. The higher the tortuosity index (T) of the vessel segment was, the greater tortuous of the vessel segment would be. The average tortuosity index of an image was 75% of tortuosity in each image. The T value of normal images was smaller than that of APROP images. When superimposing branch points, the investigators calculated the entire fundus. Compared with normal images, the APROP image showed excessive branching of vessels and therefore high segment counts. The feature vector included the average tortuosity index of each eye and the corresponding number of segments (S). The result showed that images with APROP had higher arteriolar and venular integrated curvature (IC) than those without APROP. Future studies involving quantitative features of APROP will be helfpul for its diagnosis and treatment.

Ataer-Cansizoglu *et al.* developed and evaluated the performance of a novel CBIA system named "i-ROP" for detecting plus disease and identifying the characteristics that accorded best with those described by specialists [28]. The study recruited 77 large circular cropped images. Among the images, there were 14 plus images, 16 pre-plus images, and 47 healthy images based on the experts' diagnosis. In the retinal images, the vascular structure was defined as a tree of vascular segments, where a segment was a curved structure between two connection points or between a connection point and an endpoint. Point-based measurements of arterial and venous tortuosity as well as a large circular cropped image (with a radius 6-fold longer than the disc diameter) outperformed in accuracy. The study used a Gaussian mixture model-based extraction method to quantify tortuosity. For dilation extraction, the average segment diameter and the average point diameter were calculated. Support Vector Machine classifier was used in classification of no, pre-plus or plus disease. Compared to the reference standard classifying plus and pre-plus disease, the i-ROP system achieved an accuracy of 95%. This study suggested that a well-developed automated system might perform comparably to ROP experts in the 3-level plus disease discrimination (i.e., no, pre-plus, or plus disease). In another article involving the i-ROP system, Campbell *et al.* concluded that experts in ROP appeared to focus more on features of the peripheral retina when diagnosing plus disease and take into account of tortuosity of both arteries and veins, in contrast to ICROP definitions [29].

3.2.1.1. Performance comparison

In the six methods based on classic ML, specifically in vessel detection and extraction, Gaussian matched filter had 100% recall and 95.45% precision in morphologically obtained vessel network for feature extraction in AP-ROP images. When it came to the selection of vessels, Garbor filter managed to use the MTA as the main vessel to successfully diagnose plus disease. The result showed that the sensitivity and the specificity were 89% and 95% in assessing tortuosity, respectively; the corresponding values were 73% and 66% in the measurement of vessel thickness, respectively. When diagnosing plus disease, NEAT showed a strong correlation with the golden standard of clinical evaluation of plus disease, and the Pearson's correlation coefficient was 0.802. The accuracy of MLP classifier for classifying three different categories (no, pre-plus, and plus disease) was 84.41%. On the other hand, the i-ROP system achieved 95% accuracy compared to the reference standard in classifying plus and pre-plus disease. The sensitivity for diagnosing pre-plus or worse disease was 97% and the corresponding value for plus disease was 93%. In grading pre-plus and plus disease, the ROPtool showing the pictorial scale had a good correlation with the values of the tortuosity and dilation indices. In 10 (100%) images, after the dilation of 10 blood vessels, the tortuosity index increased at all levels of dilation. In 10 images, 9/10 (90%) of the vessels were distorted differently, and the dilation index was increased.

3.2.1.2. Limitations and advantages

The most up-to-date CBIA systems were trained against a reference standard based on the diagnosis by specialists. Previous studies reported that experts have emphasized a lot on extra retinal features (such as venous tortuosity) and larger retinal fields of view during clinical diagnosis [30–32]. Therefore, these CBIA systems can emulate clinical decision better than some of the previous counterparts referring to the International Classification of

ROP (ICROP) definition in terms of plus disease diagnosis. Due to the inconsistent agreement on the extent of vascular dilation and tortuosity, there are considerable subjectivity and variability in the diagnosis at the bedside, even among experts [33]. An experiment using fundus images to diagnose plus disease unmasked that only 21% (7/34) of images were given the uniform diagnosis (with or without plus disease) by 22 experts [34]. Thus, the application of computer-aided technology to quantify and automatically distinguish the characteristics of blood vessels will effectively avoid the diagnostic error resulting from personal knowledge and experience, ultimately improving the diagnostic consistency.

Nevertheless, these systems were mostly developed to diagnose plus or pre-plus disease (in some latest studies as introduced above) without giving the description of zones and stages of ROP. Since ROP is a progressive disease, it is necessary to establish a more convenient system to help ophthalmologists to monitor the disease development. Unfortunately, other clinical symptoms related to ROP were not considered in the systems. Moreover, the analyses required the input images with a good quality. Thus, when applied clinically, there are technical and anatomical obstacles to produce gradable images. The computer system relies on the manual extraction and segmentation of the vessels, which may lower the precision of results due to possible mistakes and expert bias when selecting the target vessels, especially in the follow-up examination. A more automated system is required and may be feasible in future because of the recent advances in AI. Of note, BIO- and telemedicine-style photographs differ, given that the former are not a wide-field view of the fundus. Each of analytical methods uses an algorithm to evaluate vascular characteristics as mentioned above, consequently making comparisons among various methods both necessary and complex.

Since most of these studies were restricted to images of decent quality for computer-based analysis, manual reading to select appropriate photos is essentially unpractical. Thus, solutions must be found for the real-world application, with an algorithm based on DL to choose qualified photos, which are further discussed below.

3.2.2. Deep learning (DL) for ROP

In algorithm development, transfer learning was applied in DL to reuse a pre-trained model as a feature extractor. When dealing with image data, transfer learning is performed with predictive modeling problems. With regard to such problems, DL models that are pre-trained for large and challenging image classification tasks are frequently utilized. The researchers have subsequently developed various models for this competition. These models are improved further and expect image data as input. In the following systems using DL, ImageNet database was used in building the convolutional neural networks. On the basis of image training in ImageNet, Inception models (e.g. Inception version 1, Inception-BN, Inception-v3) were developed to predict classification. Since the network has learned highly generalizable image features from unrelated but large and highly diverse image datasets, transfer learning has been shown to improve classification performance.

3.2.2.1. i-ROP DL system

Using the DL method, Brown *et al.* developed a new system based on convolutional neural networks (CNNs) to automatically diagnose ROP [35]. The system termed “i-ROP DL” applied the DL algorithm developed by the i-ROP research team. This automated algorithm was trained with a dataset of 5511 retinal images. The reference standard diagnosis (RSD) for each image in the dataset was based on consensus of image reading by 3 experts and clinical diagnosis by 1 specialist. Two CNNs were used in the algorithms. The first one was a vessel segmentation network that was trained to export new images with the pixel value ranging from 0 to 1. The second one was trained to diagnose plus disease in the pre-processing images. Through a series of alternating filtering and down-sampling operations, the classification network simplified the image into a set of features which were subsequently converted into 3 values representing the possibility of no, pre-plus, or plus disease. During the detection of plus disease in 100 retinal images, the algorithm achieved a sensitivity and a specificity of 93% and 94%, respectively. For diagnosis of pre-plus disease, the sensitivity and the specificity were 100% and 94%, respectively. Besides, the algorithm performed better than most specialists with a quadratic-weighted κ coefficient of 0.92 when compared with the RSD. The above-mentioned study reported that the deep CNN had a comparable or even better accuracy than specialists in diagnosing plus disease. Hence, the algorithm has potential to be applied for ROP detection and monitoring in preterm babies.

Recently, Taylor *et al.* reported a score setting to quantify the severity of vascular lesion for ROP in the i-ROP DL system in order to evaluate the disease progression and regression after treatment [36]. The score setting was a quantification of disease classifications at each time point, and was then applied to monitor the disease progression. The i-ROP DL system was used for analysis of retinal images and classification of the probability of plus or pre-plus disease, or no disease. The corresponding method [(1 $\times$ probability of no disease) + (5 $\times$ probability of pre-plus disease) + (9 $\times$ probability of plus disease)] based on these 3 probabilities was subsequently utilized for a ROP vascular severity score which was assigned to an image from 1 (normal) to 9 (worst) [37]. Inter-expert variability proved to be high in diagnosing ROP according to the ICROP, and yet low in disease severity scoring [34,38]. As a result, the current ROP screening models relying on the full ICROP may be conceptualized to improve diagnosis by ranking ROP severity and quantifying image features when the detection sensitivity of clinically significant ROP (Type 1) becomes higher (i.e., $>$ 94% sensitivity) [37,39] . The study advised that postmenstrual age (PMA) ought to be considered as a factor in image-based risk prediction models because the severity score was higher in treatment-requiring (TR) group when comparing the long-term difference in the severity scores across time between TR eyes and non-TR counterparts. These findings may have significant clinical indications for disease prognosis, care delivery and management when using traditional BIO or telemedicine [40–43]. Gupta *et al.* led another parallel study of disease regression, where two i-ROP cohorts were arranged in patients treated with laser and anti-VEGF (bevacizumab) [44]. They found that the severity score was capable of monitoring disease regression after treatments, and thereby could be utilized to explore the differences between therapeutic laser-induced and anti-VEGF administration-induced regressions; a slower disease regression was observed following laser therapy.

3.2.2.2. DeepROP system

In another research, by use of deep neural networks (DNNs), Wang *et al.* developed an intelligent system called the “DeepROP” for ROP detection [45]. ROP detection in that study was divided into ROP identification and grading tasks (i.e. Id-Net and Gr-Net). A large-scale dataset comprising 7559 normal images, 1843 minor images, and 2305 severe images was used in the training. At the first step, the system was designed to identify the fundus image as “normal” or “ROP”. The ROP case would subsequently be classified into “minor” or “severe”. The “minor ROP” cases were those with the clinical features of Zone II or III at Stage 1 or 2, while the “severe ROP” cases were with the clinical features of threshold disease with Type I, II, or APROP, and Stage 4 or 5. The proposed datasets consisted of 3 parts: developing data with clinical labels to build the DNN models, data for comparison with specialists, and data collected from the external database. A large dataset was used

to train these two networks to yield the parameters. To correct the error of the system when a labeling algorithm was established after pre-training, the system calculated the confusion and error rate based on some cases evaluated by three ROP experts. The findings suggested that the DNN models could independently recognize ROP features from big data and the DeepROP might be effective and efficient in automated ROP screening.

3.2.2.3. ROP.AI system

ROP.AI is a diagnosing system of plus disease developed in Australia by use of the DL method [46]. The system applied a local database consisting of 3487 images. All images in the training set were graded by one ROP specialist and the test set was graded by another expert. The images for training were pre-processed via data augmentation to double the number of images. CNNs were used in image recognition and feature emphasis. The images were then graded and classified into normal and plus disease. The last step in developing this system was using the RMSProp optimizer to optimize the result by adjusting the operating points. After the system was established, the researchers applied 26 pre-plus images in the test set to testify the generalization and accuracy of ROP.AI, since detecting the early stage of plus disease could improve the prognosis. The system proved highly accurate in diagnosing plus disease, and was promising to be applied in quantitative analysis of ROP.

3.2.2.4. Comparison among automated systems using DL method

Since these two systems were derived from the similar DL methods, we next made a comparison between them.

The recent i-ROP DL system, developed by Taylor *et al.*, adopted a quantitative score setting of ROP from 1 (most normal) to 9 (most severe), in contrast to the previous RSD setting which was simplified into 4 categories: no ROP, mild ROP, Type 2 ROP or pre-plus disease, and Type 1 ROP [35,36] . This new RSD was based on the independent image-based diagnoses by 3 trained readers (2 ophthalmologists and 1 study coordinator) as well as the clinical diagnosis (obtained from the complete evaluation, including the dilated ophthalmoscopic examination) by a specialist [16]. The calculation involved 3 probabilities classified by the i-ROP DL system using the aforementioned method [i.e., (1 $\times$ probability of no disease) + (5 $\times$ probability of pre-plus disease) + (9 $\times$ probability of plus disease)]. On the other hand, the grading of ROP in DeepROP system was based on the clinical features of “minor ROP” (Zone II or III and Stage 1 or 2), or “severe ROP” (threshold disease, Type I, Type II, or APROP, and Stage 4, or 5). The two datasets (Id-Net and Gr-Net) were first annotated and graded by three ophthalmologists and then evaluated by one more senior clinician if the annotations of “normal” or “ROP” differed among the initial readers. The cases were excluded if the fourth ophthalmologist disagreed with the annotations.

In the study by Wang *et al.*, DNNs detected ROP upon each case rather than each image independently, whereas the CNN analysis by Brown *et al.* counted on each image in the dataset [35,45] . Considering that Wang *et al.* didn’t describe the fields of view of images whereas Brown *et al.* demonstrated that only single posterior-pole fundus images were analyzed, it seemed that the DeepROP might analyze different fields of view from multiple images captured by Retcam for each case. Notably, the diagnosis was based not only on the arterial tortuosity, but also the venous dilation of posterior retinal vessels that was used to diagnose plus disease when comparing to a standard retinal photograph in 1980s [47,48]. Thus, the reason why Brown and colleagues analyzed only the posterior images was probably that i-ROP DL was originally designed to detect 3-tier grading classification of plus disease rather than grading of ROP, despite that the other system established by Taylor *et al.* has applied the algorithm to quantitatively score ROP.

Unlike the other two systems, ROP.AI has a training set and a test set that are only diagnosed by one expert separately. Therefore, this may restrict the applicability of the system since the high inter-expert variability in diagnosing plus disease is known. The ROP.AI system has covered a population of major races (European and Maori) in Australia and New Zealand, while the DeepROP system has a database involving Asian people. These two databases may offset the ethnic heterogeneity and improve the generalization of diagnosing systems if a shared image database is established. ROP.AI was developed only to classify normal and plus disease in the initial development and applied a small number of pre-plus images in the test set. On the other hand, i-ROP DL was trained using images in three classifications (normal, pre-plus, plus). It seemed that i-ROP DL had a higher confidence in detecting pre-plus disease.

Concerning the performance of the aforementioned automated systems, i-ROP DL achieved a sensitivity and a specificity of 93% and 94% in diagnosing plus disease, respectively. For the diagnosis of pre-plus disease, the sensitivity and the specificity were 100% and 94%, respectively. On the other hand, DeepROP showed that Id-Net (ROP and normal) attained a sensitivity and a specificity of 96.64% and 99.33%, respectively, whereas the corresponding values for grading of ROP (minor and severe) were 88.46% and 92.31%. Both systems performed better than at least one expert in terms of accuracy. ROP.AI attained a sensitivity and a specificity of 96.6% and 98.0%, respectively.

3.2.2.5. DL for evaluation of image quality

Since the image analysis systems described above were developed and tested on the retinal photographs with proper quality, the generalization of these AI systems is suspicious because photographs taken by different operators may not be necessarily qualified. Similarly, some of the studies in telemedicine applications put forward that poor image quality may exert adverse effects on diagnostic accuracy and confidence [7,49–51]. Therefore, in the real-world clinical applications, images are meaningless if their quality is not high enough. Recently, Coyner *et al.* developed an algorithm based on CNNs to determine the quality of fundus images in ROP, which may serve as a pre-screening tool for telemedicine applications and ROP diagnostic techniques [52]. The accuracy of the test set was 89.1%, with the area under ROC equal to 0.964, and the area under the precision-recall curve equal to 0.966. A notable limitation of this model was that only images of posterior-pole fundus were included for training and testing, which is the same as the i-ROP and the i-ROP DL systems, even if it proved less effective than a 5-image set in the study by Roger *et al.* Thus, the authors suggested that CNNs were able to provide a strong discrimination between low- and high-quality images and be generalized to all clinical fields involving image-based diagnosis since the threshold of the CNN-labeled images could be adjusted in an application-dependent manner with no apparent adverse influence.

3.2.3. Multiple instance learning (MIL) for ROP

Rani *et al.* used MIL methods for ROP detection [53]. In MIL, the images were split into a grid of rectangular patches and features were extracted from each patch. In the end, all the patches belonging to the same image were grouped together and termed as a bag. The bag was given a label depending on whether it was disease-affected or healthy. In detail, the dataset comprised 10 ROP and 10 normal images. First, the images were converting into a grid of patches. Second, the set of features were extracted from each patch, where the set of features equaled to 1 instance. Third, each instance was given a label, and all instances with the same label consisted of a group. Fourth, for a disease image, the bag label was defined as +1, whereas for a normal image the bag label

was -1. Finally, the miGraph classifier processed all training bags by deriving affinity matrices and defined an efficient graph kernel regarding the clique information. The images were therefore classified into ROP or normal. The finding suggested that the accuracy of diagnosis was 83.33%, with the values of sensitivity and specificity 100% and 71.43%, respectively; the AUC-ROC was 0.63. The research demonstrated that MIL was effective in detecting ROP images. Still, further research in this direction is needed since the related reports remain scarce. Additionally, MIL methods can be another direction to build an automated system for ROP detection in the future apart from DL.

3.2.4. Characteristics of datasets

The methods described above applied 20 to 11,707 images in developing the algorithm. Within each dataset, multiple expert ophthalmologists confirmed the diagnosis of ROP from which the algorithms learned to output results. In detail, the i-ROP researchers created a reference standard diagnosis to the images according to the major agreement of three specialists in the training dataset. Next, the study group of i-ROP developed an automated diagnosing system with DL (i.e. i-ROP DL), where three image readers, together with one clinician who gave a clinical diagnosis based on the dilated ophthalmic examination, graded the images independently. The training set in the DeepROP system was also graded by a four-member expert group with a similar team structure. However, unlike the i-ROP DL system, no additional evaluation from the fourth clinician would be required unless the initial labels by the three experts were inconsistent. Under such circumstances, all the four experts would revisit the medical records, ultimately drawing an overall conclusion. The other aforementioned methods developed reference diagnosis based on the consensus of 1 to 3 ROP experts, using a similar procedure of the i-ROP system. Besides, the source of databases may vary. For instance, telemedicine databases were established and frequently employed in some methods. Fundus images from the Telemedicine for ROP In Calgary (TROPIC) database were assessed with Garbor filters, while those from Auckland Regional Telemedicine ROP (ART-ROP) image library were determined with ROP.AI. Notably, methods using DL seem to prefer a multicenter database for training and testing, because the capacity of DL allows the input from a large-scale dataset covering various populations. In contrast, methods applying CML fit better to a relatively small-scale dataset since they were developed by manual feature extraction. Nevertheless, all the images included in datasets for either methods mentioned above should have a high quality. Interestingly, not all of the previous studies involved normal retinal images in the training set. Methods such as ROPtool, NEAT and MLP classifiers which were designed to discriminate plus or pre-plus disease from non-plus healthy controls might need a preliminary assessment by clinicians prior to the computer diagnosis. Hence, they are somehow less compatible with the telemedicine program for initial screening.

3.2.5. Validation of results

Algorithm validation is an important step ensuring that the algorithms could perform well in a clinical setting and mimic the real situation. The validated algorithms are expected to provide reliable results to clinical doctors in choosing a suitable system. There are ways to perform validation. Cross-validation was used to compute the error and accuracy of the algorithms. For example, Rajashekaar *et al.* applied cross-validation to receive the classification error rate of the features calculated according to the T value and segment count. In addition, Ataer-Cansizoglu *et al.* and Pour *et al.* performed a k-fold cross-validation procedure to evaluate the accuracy of the system by examining the trained data. The result showed that i-ROP system achieved an accuracy of 95%.

Studies using the DL method applied a clinical set of images to test the performance of the system *versus* experts. Brown *et al.* used a test dataset that was never applied in the training or verification image collection comprising 100 images (54 normal, 31 plus and 15 pre-plus cases). The automated diagnosis by the i-ROP DL system of the test dataset was compared with that by 8 ROP experts. The algorithm was evaluated by the 5-fold cross-validation and tested on an independent set of 100 images. The i-ROP was validated to diagnose 91% of images correctly. On the other hand, the system by Wang *et al.* was applied clinically in 6 different hospitals in Sichuan, China, for ROP screening. A well-designed clinical setting was established through 3 steps that tested the DeepROP system as follows: First, retinal fundus images were uploaded. Second, the automated ROP detection was started to obtain the initial diagnosis. Third, the results were graded by specialists to get the final diagnosis, provided that the confidence score for the automated ROP detection platform was below a threshold. This final diagnostic report was regarded as the true diagnosis and used as the RSD to evaluate the accuracy of the algorithm in the clinical setting. Based on the results, the DeepROP displayed 94.2% accuracy in diagnosing and classifying ROP. The developer of the ROP.AI applied both internal and external validation to assess the ability to diagnose plus disease. The training set was split into a ratio of 80:20 for only training and a hold-out set for internal test, respectively. For further evaluation, an external test that comprised 116 images was conducted. The overall internal and external accuracies were 97.3% and 85.6%, respectively.

In the system using multiple instance learning, Rani *et al.* ran a robust validation, where the dataset comprised 28 ROP-affected and 28 healthy images. In the validation, the demarcation line, ridge, extraretinal fibrovascular proliferation, and detached retina were localized predominately in the vasculature of the retina. These four features were summarized from each image and the bag containing patches from the same image was fed into two MIL classifiers, i.e. miGraph and citation-kNN. It was reported that miGraph outperformed citation-kNN. These classifiers categorized images into ROP-affected and healthy ones. After validation of the algorithms, the systems both performed well-fitting in the clinical setting. There was a good agreement between the training and the testing, indicating the capability of these algorithms to be applicable in the clinical setting. In this work, MIL had an accuracy greater than 95% in diagnosing ROP.

4. Discussion

4.1. Future perspectives

Due to the growing shortage of pediatric ophthalmologists, incorporating telemedicine and CBIA systems for screening and examination of ROP is urgently required all over the world. This can improve the efficiency and effectiveness of detection, supervision and in-time treatment of ROP for the preterm babies. Even though the current automated systems perform well in the research, there have been few reports about the accuracy of ROP on large-scale clinical applications. Moreover, it is difficult to compare the existing systems in a uniform way due to the lack of a standard. Hence, the clinical accuracy of these systems in real-world is unknown. A large scale of clinical trials for the application of these systems is needed not only for verifying the true accuracy but also for building a large database. In addition, the development of databases and clinical trials should rely on a more variable population with different genders and races throughout the world. Therefore, it is called upon to establish a program to compare the existing ROP programs, like the Retinal Fundus Glaucoma Challenge

and REFUGE held in conjunction with MICCAI 2018 [54]. While it may be hard to conduct the above-mentioned comparison under the current circumstances due to the relatively small population of infants, it would be more feasible considering the rapid progresses in medicine and technology. From a current view, an incorporation of telemedicine and computer-based image analysis can provide detection, supervision and in-time treatment of ROP for the preterm babies. In the future, innovative programs are expected to be designed based on various datasets with the population from different countries so that the networks can be modified according to specific situations.

Declaration of Competing Interest

The authors declare no competing financial interests.

Acknowledgements

The study is supported by the National NSFC (82074169, 82071372), Hygiene & Health Appropriated Technology and Promoting Project of Guangdong Province (202006130025341204, 201905270933056876), the Outstanding Scholar Program of Bioland Laboratory (Guangzhou Regenerative Medicine and Health Guangdong Laboratory) (2018GZR110102002) and Science and Technology Program of Guangzhou (202007030012).

Supplementary materials

Supplementary material associated with this article can be found, in the online version, at doi:10.1016/j.cmpb.2020.105871.

References

[1] A. Hellström, L.E. Smith, O. Dammann, in: Retinopathy of prematurity, 382, Lancet (London, England), 2013, pp. 1445–1457.
[2] W.V. Good, R.J. Hardy, V. Dobson, E.A. Palmer, D.L. Phelps, B. Tung, M. Redford, Final visual acuity results in the early treatment for retinopathy of prematurity study, Archives of ophthalmology (Chicago, Ill.: 1960 128 (2010) 663–671.
[3] M.D. Abramoff, P.T. Lavin, M. Birch, N. Shah, J.C. Folk, Pivotal trial of an autonomous AI-based diagnostic system for detection of diabetic retinopathy in primary care offices, NPJ Digit Med 1 (2018) 39.
[4] W.M. Fierson, O. American Academy of Pediatrics Section on, O. American Academy of, O. American Association for Pediatric, Strabismus, O. American Association of Certified, Screening examination of premature infants for retinopathy of prematurity, Pediatrics 131 (2013) 189–195.
[5] C. Wu, R.A. Petersen, D.K. VanderVeen, RetCam imaging for retinopathy of prematurity screening, J AAPOS 10 (2006) 107–111.
[6] M.F. Chiang, J.D. Keenan, J. Starren, Y.E. Du, W.M. Schiff, G.R. Barile, J. Li, R.A. Johnson, D.J. Hess, J.T. Flynn, Accuracy and reliability of remote retinopathy of prematurity diagnosis, Arch Ophthalmol 124 (2006) 322–327.
[7] M.F. Chiang, L. Wang, M. Busuioc, Y.E. Du, P. Chan, S.A. Kane, T.C. Lee, D.J. Weissgold, A.M. Berrocal, O. Coki, J.T. Flynn, J. Starren, Telemedical retinopathy of prematurity diagnosis: accuracy, reliability, and image quality, Arch Ophthalmol 125 (2007) 1531–1538.
[8] S.K. Wang, N.F. Callaway, M.B. Wallenstein, M.T. Henderson, T. Leng, D.M. Moshfeghi, SUNDROP: six years of screening for retinopathy of prematurity with telemedicine, Can J Ophthalmol 50 (2015) 101–106.
[9] A.L. Ells, J.M. Holmes, W.F. Astle, G. Williams, D.A. Leske, M. Fielden, B. Uphill, P. Jennett, M. Hebert, Telemedicine approach to screening for severe retinopathy of prematurity: a pilot study, Ophthalmology 110 (2003) 2113–2117.
[10] C. Dhaliwal, E. Wright, C. Graham, N. McIntosh, B.W. Fleck, Wide-field digital retinal imaging versus binocular indirect ophthalmoscopy for retinopathy of prematurity screening: a two-observer prospective, randomised comparison, Br J Ophthalmol 93 (2009) 355–359.
[11] W.M. Fierson, A. Capone, Jr., O. American Academy of Pediatrics Section on, A.A.o.C.O. American Academy of Ophthalmology, Telemedicine for evaluation of retinopathy of prematurity, Pediatrics 135 (2015) e238–e254.
[12] D. Rajashekar, G. Srinivasa, A. Vinekar, Comprehensive Retinal Image Analysis for Aggressive Posterior Retinopathy of Prematurity, PLoS One 11 (2016) e0163923.
[13] P.K. Shah, A. Ramya, V. Narendran, Telemedicine for ROP, Asia Pac J Ophthalmol (Phila) 7 (2018) 52–55.
[14] A. Thanos, Y. Yonekawa, B. Todorich, D.M. Moshfeghi, M.T. Trese, Screening and treatments using telemedicine in retinopathy of prematurity, Eye Brain 8 (2016) 147–151.
[15] G.E. Quinn, A. Ells, A. Capone, Jr., G.B. Hubbard, E. Daniel, P.L. Hildebrand, G.S. Ying, R.O.P.C.G. e, Analysis of Discrepancy Between Diagnostic Clinical Examination Findings and Corresponding Evaluation of Digital Images in the Telemedicine Approaches to Evaluating Acute-Phase Retinopathy of Prematurity Study, JAMA Ophthalmol 134 (2016) 1263–1270.
[16] M.C. Ryan, S. Ostmo, K. Jonas, A. Berrocal, K. Drenser, J. Horowitz, T.C. Lee, C. Simmons, M.A. Martinez-Castellanos, R.V. Chan, M.F. Chiang, Development and Evaluation of Reference Standards for Image-based Telemedicine Diagnosis and Clinical Research Studies in Ophthalmology, AMIA Annu Symp Proc, 2014 (2014) 1902–1910.
[17] G.M. Richter, S.L. Williams, J. Starren, J.T. Flynn, M.F. Chiang, Telemedicine for retinopathy of prematurity diagnosis: evaluation and challenges, Surv Ophthalmol 54 (2009) 671–685.
[18] D.K. Wallace, S.F. Freedman, Z. Zhao, S.H. Jung, Accuracy of ROPtool vs individual examiners in assessing retinal vascular tortuosity, Arch Ophthalmol 125 (2007) 1523–1530.
[19] A.E. Kiely, D.K. Wallace, S.F. Freedman, Z. Zhao, Computer-assisted measurement of retinal vascular width and tortuosity in retinopathy of prematurity, Arch Ophthalmol 128 (2010) 847–852.
[20] D.K. Wallace, Z. Zhao, S.F. Freedman, A pilot study using "ROPtool" to quantify plus disease in retinopathy of prematurity, J AAPOS 11 (2007) 381–387.
[21] D.A. Nasrazadani, D.K. Wallace, S.F. Freedman, S.G. Prakalapakorn, Development of a scale for grading pre-plus and plus disease using retinal images: A pilot study, J AAPOS 22 (2018) 316–319.
[22] E.K. Pour, H. Pourreza, K.A. Zamani, A. Mahmoudi, A.M.M. Sadeghi, M. Shadravan, R. Karkhaneh, R.R. Pour, M.R. Esfahani, Retinopathy of Prematurity-assist: Novel Software for Detecting Plus Disease, Korean J Ophthalmol 31 (2017) 524–532.
[23] K. Balaskas, S. Tiew, G. Czanner, A.L. Tan, J. Ashworth, S. Biswas, T. Aslam, The Novel Evidenced Assessment of Tortuosity system: interobserver reliability and agreement with clinical assessment, Acta Ophthalmol 94 (2016) e421–e426.
[24] C.M. Wilson, K. Wong, J. Ng, K.D. Cocker, A.L. Ells, A.R. Fielder, Digital image analysis in retinopathy of prematurity: a comparison of vessel selection methods, J AAPOS 16 (2012) 223–228.
[25] C.M. Wilson, A.L. Ells, A.R. Fielder, The challenge of screening for retinopathy of prematurity, Clin Perinatol 40 (2013) 241–259.
[26] F. Oloumi, R.M. Rangayyan, P. Casti, A.L. Ells, Computer-aided diagnosis of plus disease via measurement of vessel thickness in retinal fundus images of preterm infants, Comput Biol Med 66 (2015) 316–329.
[27] F. Oloumi, R.M. Rangayyan, A.L. Ells, Computer-aided diagnosis of plus disease in retinal fundus images of preterm infants via measurement of vessel tortuosity, Conf Proc IEEE Eng Med Biol Soc, 2015 (2015) 4338–4342.
[28] E. Ataer-Cansizoglu, V. Bolon-Canedo, J.P. Campbell, A. Bozkurt, D. Erdogmus, J. Kalpathy-Cramer, S. Patel, K. Jonas, R.V. Chan, S. Ostmo, M.F. Chiang, R.O.P.R.C. i, Computer-Based Image Analysis for Plus Disease Diagnosis in Retinopathy of Prematurity: Performance of the "i-ROP" System and Image Features Associated With Expert Diagnosis, Transl Vis Sci Technol 4 (2015) 5.
[29] J.P. Campbell, E. Ataer-Cansizoglu, V. Bolon-Canedo, A. Bozkurt, D. Erdogmus, J. Kalpathy-Cramer, S.N. Patel, J.D. Reynolds, J. Horowitz, K. Hutcheson, M. Shapiro, M.X. Repka, P. Ferrone, K. Drenser, M.A. Martinez-Castellanos, S. Ostmo, K. Jonas, R.V. Chan, M.F. Chiang, R.O.P.R.C. Imaging, Informatics in, Expert Diagnosis of Plus Disease in Retinopathy of Prematurity From Computer-Based Image Analysis, JAMA Ophthalmol 134 (2016) 651–657.
[30] R. Rao, N.J. Jonsson, C. Ventura, R. Gelman, M.A. Lindquist, D.S. Casper, M.F. Chiang, Plus disease in retinopathy of prematurity: diagnostic impact of field of view, Retina 32 (2012) 1148–1155.
[31] N.J. Hewing, D.R. Kaufman, R.V. Chan, M.F. Chiang, Plus disease in retinopathy of prematurity: qualitative analysis of diagnostic process by experts, JAMA Ophthalmol 131 (2013) 1026–1032.
[32] K.M. Keck, J. Kalpathy-Cramer, E. Ataer-Cansizoglu, S. You, D. Erdogmus, M.F. Chiang, Plus disease diagnosis in retinopathy of prematurity: vascular tortuosity as a function of distance from optic disk, Retina 33 (2013) 1700–1707.
[33] L. Ghergherehchi, S.J. Kim, J.P. Campbell, S. Ostmo, R.V.P. Chan, M.F. Chiang, Plus Disease in Retinopathy of Prematurity: More Than Meets the ICROP? Asia Pac J Ophthalmol (Phila) 7 (2018) 152–155.
[34] M.F. Chiang, L. Jiang, R. Gelman, Y.E. Du, J.T. Flynn, Interexpert agreement of plus disease diagnosis in retinopathy of prematurity, Arch Ophthalmol 125 (2007) 875–880.
[35] J.M. Brown, J.P. Campbell, A. Beers, K. Chang, S. Ostmo, R.V.P. Chan, J. Dy, D. Erdogmus, S. Ioannidis, J. Kalpathy-Cramer, M.F. Chiang, C. Imaging, Informatics in Retinopathy of Prematurity Research, Automated Diagnosis of Plus Disease in Retinopathy of Prematurity Using Deep Convolutional Neural Networks, JAMA Ophthalmol 136 (2018) 803–810.
[36] S. Taylor, J.M. Brown, K. Gupta, J.P. Campbell, S. Ostmo, R.V.P. Chan, J. Dy, D. Erdogmus, S. Ioannidis, S.J. Kim, J. Kalpathy-Cramer, M.F. Chiang, C. Imaging, Informatics in Retinopathy of Prematurity, Monitoring Disease Progression With a Quantitative Severity Scale for Retinopathy of Prematurity Using Deep Learning, JAMA Ophthalmol, 2019.
[37] T.K. Redd, J.P. Campbell, J.M. Brown, S.J. Kim, S. Ostmo, R.V.P. Chan, J. Dy, D. Erdogmus, S. Ioannidis, J. Kalpathy-Cramer, M.F. Chiang, C. Imaging, Informatics in Retinopathy of Prematurity Research, Evaluation of a deep learning image assessment system for detecting severe retinopathy of prematurity, Br J Ophthalmol, 2018.
[38] J. Kalpathy-Cramer, J.P. Campbell, D. Erdogmus, P. Tian, D. Kedarisetti, C. Moleta, J.D. Reynolds, K. Hutcheson, M.J. Shapiro, M.X. Repka, P. Ferrone, K. Drenser, J. Horowitz, K. Sonmez, R. Swan, S. Ostmo, K.E. Jonas, R.V. Chan,

M.F. Chiang, C. Imaging, Informatics in Retinopathy of Prematurity Research, Plus Disease in Retinopathy of Prematurity: Improving Diagnosis by Ranking Disease Severity and Using Quantitative Image Analysis, Ophthalmology 123 (2016) 2345–2351.
[39] J.P. Campbell, J. Kalpathy-Cramer, D. Erdogmus, P. Tian, D. Kedarisetti, C. Moleta, J.D. Reynolds, K. Hutcheson, M.J. Shapiro, M.X. Repka, P. Ferrone, K. Drenser, J. Horowitz, K. Sonmez, R. Swan, S. Ostmo, K.E. Jonas, R.V.P. Chan, M.F. Chiang, I.I. Retinopathy, Plus Disease in Retinopathy of Prematurity A Continuous Spectrum of Vascular Abnormality as a Basis of Diagnostic Variability, Ophthalmology 123 (2016) 2338–2344.
[40] M.F. Chiang, M. Melia, A.N. Buffenn, S.R. Lambert, F.M. Recchia, J.L. Simpson, M.B. Yang, Detection of Clinically Significant Retinopathy of Prematurity Using Wide-angle Digital Retinal Photography A Report by the American Academy of Ophthalmology, Ophthalmology 119 (2012) 1272–1280.
[41] D.T. Weaver, Telemedicine for retinopathy of prematurity, Curr Opin Ophthalmol 24 (2013) 425–431.
[42] G.E. Quinn, G.S. Ying, E. Daniel, P.L. Hildebrand, A. Ells, A. Baumritter, A.R. Kemper, E.B. Schron, K. Wade, R.O.P.C.G. e, Validity of a telemedicine system for the evaluation of acute-phase retinopathy of prematurity, JAMA Ophthalmol 132 (2014) 1178–1184.
[43] W.M. Fierson, A.A.O.P.S.o. Ophthalmology, O. American Academy Of, O. American Association For Pediatric, Strabismus, O. American Association Of Certified, Screening Examination of Premature Infants for Retinopathy of Prematurity, Pediatrics (2018) 142.
[44] K. Gupta, J.P. Campbell, S. Taylor, J.M. Brown, S. Ostmo, R.V.P. Chan, J. Dy, D. Erdogmus, S. Ioannidis, J. Kalpathy-Cramer, S.J. Kim, M.F. Chiang, C. Imaging, Informatics in Retinopathy of Prematurity, A Quantitative Severity Scale for Retinopathy of Prematurity Using Deep Learning to Monitor Disease Regression After Treatment, JAMA Ophthalmol, 2019.
[45] J. Wang, R. Ju, Y. Chen, L. Zhang, J. Hu, Y. Wu, W. Dong, J. Zhong, Z. Yi, Automated retinopathy of prematurity screening using deep neural networks, EBioMedicine 35 (2018) 361–368.
[46] Z. Tan, S. Simkin, C. Lai, S. Dai, Deep Learning Algorithm for Automated Diagnosis of Retinopathy of Prematurity Plus Disease, Translational vision science & technology 8 (2019) 23.
[47] C.f.R.o.P.C. Group, Multicenter trial of cryotherapy for retinopathy of prematurity. Preliminary results. Cryotherapy for Retinopathy of Prematurity Cooperative Group, Archives of ophthalmology (Chicago, Ill.: 1960) 106 (1988) 471–479.
[48] G. Early Treatment For Retinopathy Of Prematurity Cooperative, Revised indications for the treatment of retinopathy of prematurity: results of the early treatment for retinopathy of prematurity randomized trial, Arch Ophthalmol 121 (2003) 1684–1694.
[49] R. Briggs, J.E. Bailey, C. Eddy, I. Sun, A methodologic issue for ophthalmic telemedicine: image quality and its effect on diagnostic accuracy and confidence, J Am Optom Assoc 69 (1998) 601–605.
[50] M.F. Chiang, R. Gelman, M.E. Martinez-Perez, Y.E. Du, D.S. Casper, L.M. Currie, P.D. Shah, J. Starren, J.T. Flynn, Image analysis for retinopathy of prematurity diagnosis, J AAPOS 13 (2009) 438–445.
[51] T. Lundberg, G. Westman, S. Hellstrom, H. Sandstrom, Digital imaging and telemedicine as a tool for studying inflammatory conditions in the middle ear–evaluation of image quality and agreement between examiners, Int J Pediatr Otorhinolaryngol 72 (2008) 73–79.
[52] A.S. Coyner, R. Swan, J.M. Brown, J. Kalpathy-Cramer, S.J. Kim, J.P. Campbell, K.E. Jonas, S. Ostmo, R.V.P. Chan, M.F. Chiang, Deep Learning for Image Quality Assessment of Fundus Images in Retinopathy of Prematurity, AMIA Annu Symp Proc 2018 (2018) 1224–1232.
[53] P. Rani, R. ElagiriRamalingam, K.T. Rajamani, M. Kandemir, D. Singh, in: Multiple Instance Learning: Robust Validation on Retinopathy of Prematurity, 9, International Journal of Control Theory and Applications (IJCTA) (International Sciences Press), 2016, pp. 451–459.
[54] J.I. Orlando, H. Fu, J. Barbosa Breda, K. van Keer, D.R. Bathula, A. Diaz-Pinto, R. Fang, P.A. Heng, J. Kim, J. Lee, J. Lee, X. Li, P. Liu, S. Lu, B. Murugesan, V. Naranjo, S.S.R. Phaye, S.M. Shankaranarayana, A. Sikka, J. Son, A. van den Hengel, S. Wang, J. Wu, Z. Wu, G. Xu, Y. Xu, P. Yin, F. Li, X. Zhang, Y. Xu, H. Bogunovic, REFUGE Challenge, A unified framework for evaluating automated methods for glaucoma assessment from fundus photographs, Med Image Anal 59 (2020) 101570.